

Mohammad Adab

📍 NIT Srinagar, J&K, India · 📞 +91-6006711641 · 📁 Portfolio · ✉ Email · 🔗 LinkedIn · 🐙 GitHub

EXPERIENCE

Software Development & Research Assistant · GAASH Lab, NIT Srinagar

Feb 2026 – Present

- Built scalable Django and PostgreSQL solutions, including an ERP system for **500+ scholars** and a DRF-based admission platform for **4000+ applicants**, leveraging RESTful APIs and automated form workflows to streamline institutional operations.
- Conducted **deep learning** and **computer vision** research, accelerating model deployment and evaluation to guide ongoing lab initiatives.

Frontend Engineer Intern · Traxevo Web Limited

May 2026 – Present

- Reduced page load time by **40%** for **15,000+** monthly users by engineering high-performance React.js interfaces with optimized Hooks, lazy loading, and REST API integration.
- Increased development efficiency by **30%** by designing a reusable component library and streamlining Agile sprint cycles while maintaining cross-browser compatibility.

Computer Vision & Deep Learning Intern · IIT Roorkee

Dec 2025 – Mar 2026

- Designed a hybrid object detection architecture combining a ResNet-50 backbone with Vision Transformer (ViT) attention layers to maximize feature representation.
- Achieved **0.813 mAP@0.5** and **88.4% precision**, outperforming the YOLOv5 baseline by **+5.8% mAP** and **+4.2% F1-score**.

PROJECTS

Airline Fuel Optimization Agent 🤖 | Python, Ollama, Docker, MCP, Strands SDK, LLMs, Optimization Algorithms

- Built an AI agent to optimize flight fuel consumption by analyzing flight plans and live weather data for route and altitude recommendations.
- Designed a modular multi-agent architecture using **AWS Strands SDK** and **Model Context Protocol (MCP)** for plug-and-play tool integration.
- Integrated real-time **METAR/TAF weather data** via **MCP tools**, building a live **ETL** pipeline to dynamically refine routing decisions based on aviation conditions.
- Containerized the application with **Docker** supporting both local LLMs (**Ollama**) and **AWS Bedrock**, achieving a **1% fuel optimization** in sample scenarios, following **secure coding** and **cloud-portable** deployment practices.

ReviewFlow AI - Agentic Customer Review Intelligence System 🤖 | Python, LangChain, LangGraph, FLAN-T5, LCEL, SMTP

- Built an end-to-end AI system using **LangChain** and **LangGraph** with multi-agent orchestration, reducing manual review triage by **~80%** with **<15s response times**.
- Engineered a stateful multi-node **RAG-ready** pipeline using LCEL chains with prompt engineering on **FLAN-T5**, achieving **~90% sentiment classification accuracy**.
- Automated real-time escalation of negative reviews via Gmail **SMTP**, **SDK integration**, triggering stakeholder alerts within **2 seconds**.

Hydra - Distributed MapReduce Engine 🤖 | Python, FastAPI, REST, Multithreading, React, Distributed Systems

- Built a **fault-tolerant distributed MapReduce** engine on a **Master/Worker** model, scheduling parallel map and reduce tasks across independent **OS-process workers** over a **REST/HTTP** control plane with thread-safe **concurrent** coordination.
- Implemented **heartbeat-based failure detection** with automatic **task reassignment**, lost-output recomputation, and **idempotent stale-result rejection**, guaranteeing correct **distributed key-value aggregation** with **zero data loss** under mid-job worker failure.
- Engineered a **file-based shuffle** with deterministic **CRC32 hash partitioning** and a real-time **React dashboard** monitoring **job lifecycle**, **worker health**, and live cluster events.

TECHNICAL SKILLS

Languages: C, Java, Python, JavaScript

Frameworks & Libraries: React.js, Django, FastAPI, Pandas, Scikit-learn, LangChain, LangGraph, LCEL, STrands SDK, Ollama, PyTorch

Databases: SQL, PostgreSQL, ChromaDB, FAISS, Celery, Redis, ETL

Tools & Platforms: Git, GitHub, GitHub Actions, CI/CD, Docker, Nginx, AWS Bedrock, Postman, MCP

AI/ML: Machine Learning, Deep Learning, NLP, Computer Vision, Neural Networks, Model Evaluation, Model Optimization, RAG, GenAI, Agentic AI, Prompt Engineerin, Multi-Agent Orchestration, LLM Fine-Tuning, SDK Integration

EDUCATION

National Institute of Technology Srinagar

Aug 2023 – Present

B.Tech in Information Technology

CGPA: 7.99/10

Relevant Courses: Machine Learning, OOPs, DBMS, DSA, Operating Systems, DBMS, Computer Networks